

An Algorithmic Approach to 4-way Autonomous Intersection Management

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Abstract—The objective of this research is to present an approach focused on optimizing intersection management, a subset of end-to-end autonomous driving. The proposed method relies on vehicle-to-vehicle (V2V) communication and an intersection signal amplifier to enhance intersection handling. This approach prioritizes improving intersection management by using V2V communication in place of conventional traffic signal intersections and also increases intersection safety while streamlining resource allocation and time efficiency. Evaluations are conducted using the microscopic traffic simulator SUMO (Simulation of Urban MObility), and the results indicate a noticeable reduction in congestion. The intended goal here is to reduce the average time taken for travel and provide travel convenience by aiming for constant speed. Therefore provides an efficient algorithm for implementing and testing autonomous intersection management strategies with autonomous driving systems, contributing to developing more efficient and safer autonomous transportation solutions.

Index Terms—V2V communication, Autonomous Intersection Management, Connected Autonomous Vehicles.

I. INTRODUCTION

The significance of intersections in traffic management is evident with uncontrolled intersections contributing to a substantial percentage of road traffic crashes in India, as indicated by IATSS research [1]. Conventional traffic light installations at every junction can be costly and inefficient due to varying traffic loads. Moreover, the future of traffic management is evolving toward Connected Autonomous Vehicles (CAVs), which necessitates new strategies.

Different autonomous vehicle manufacturers will implement various navigation strategies, typically controlled centrally. While this suits scenes that do not require coordination, intersections pose a unique challenge. Traditional signalized controllers are ineffective, and the true potential of CAVs

remains untapped [2]. Coordinating CAVs poses privacy and data-sharing challenges among autonomous service providers discussed in [3]. To address this, we propose a decentralized approach where vehicles share minimal collision-related data only with relevant peers. This approach ensures efficiency, robustness, and privacy preservation without relying on a central point of failure.

Unlike typical autonomous intersection management (AIM) strategies that rely on power-intensive Intersection Management (IM) units discussed in [4], our approach transforms IM units into signal amplification hubs, not computational centers. It is decentralized, requiring fewer computational resources by making local decisions, and ensuring real-time responsiveness in congested traffic scenarios. Approaching an intersection, a vehicle initially relies on local information for a preliminary decision, fetching traffic data when within communication range, and adjusting its speed for safe entry without exceeding the designated arrival time.

Our research offers a resource-efficient and cost-effective template for developers testing new autonomous vehicle features, addressing the challenges posed by a real-world problem. In comparison with previous work [12], our system accommodates greedy vehicle behavior, considered in global decision-making. The global decision algorithm is uniform across all vehicles, with the IM unit serving as an amplification medium. Specific communication methodologies and error handling are beyond our scope, but our system remains functional even if a node becomes dysfunctional.

II. RELATED WORK

For intersection management, some techniques aim to solve traffic congestion problems through reservation-based systems [5]. In [6], authors propose a multi-agent approach in which vehicles follow predefined road rules which are the optimal solution to the traffic flow problem. Finally, in [7] and [8] a single agent is trained to handle unsignalized intersections using Deep Q-Networks (DQNs) [9]; in such works, the environment is populated by traffic cars that follow the Intelligent Driver Model (IDM) [10] to control their speeds, while the vehicle trained using this chooses a discrete action in a correct time step to avoid a collision with incoming traffic. Another approach which is based on end-to-end driving using Deep Reinforcement Learning (DRL) is demonstrated in [11], where a lane following policy was obtained by training a neural network that receives a single image as input; however, such a model is suited to solve simple tasks and it does not consider the interaction with other vehicles in the surroundings since it is trained in an obstacle-free environment. One more method is based on a supervised approach as Imitation and Conditional Imitation Learning; nevertheless, these techniques have the main limitation of needing a huge amount of real data to avoid over-fitting. And they work only in the same traffic patterns i.e., are not dynamic. The preliminary idea [12] for our research considers vehicle-to-infrastructure (V2I) communication where all vehicles share their data with a centralized controller which here is the IM unit. It collects data from all vehicles makes decisions and shares it with them. The vehicles are bound to follow the decision made by this unit. This method however does not allow the freedom for each vehicle to request for desired priority and the IM unit acts as a single point of failure. If it goes down, the entire system goes down. Our approach differs from this being V2V communication and each vehicle can assign themselves higher priority which will be taken into consideration while making global decisions. In [13] a State Feedback Controller is proposed for the traffic light system control at the four-leg intersection which is then simulated using SUMO. To achieve AIM, [14] employs a multi-agent management system comprising Vehicle Agents (VAs) and an Intersection Agent (IA). In this approach, vehicles share their anticipated intersection entry points and estimated arrival times. Simultaneously, the IA coordinates reservations within a time-space framework and communicates the reservation outcomes to the vehicles. Armed with this information, individual VAs optimize their speed, which may involve speed reduction or efficient braking strategies, well in advance of approaching the intersection area. This proactive adjustment enables them to pass through the intersection smoothly, minimizing the need for complete stops and ensuring efficient traversal.

III. PROPOSED SYSTEM

A. Intersection environment settings

Numerous types of intersections exist, but our focus in this study is on the commonly encountered three-lane, four-way intersection, also known as a crossroad, see Fig. 1 and 2. Our strategy can be easily expanded to accommodate other intersection shapes by adjusting the relevant entries in the conflict list. The intersection consists of a total of 12 incoming and outgoing lanes, each identified by a unique ID r_i (where $i = 0, 1, \dots, 11$). The lanes are numbered in a clockwise manner, starting from the topmost lane. It is important to note that each lane has a single outgoing lane and direction, simplifying the analysis and modifications as the movement of vehicles within the intersection area or at any given point can be uniquely represented by the corresponding incoming and intended outgoing lanes.

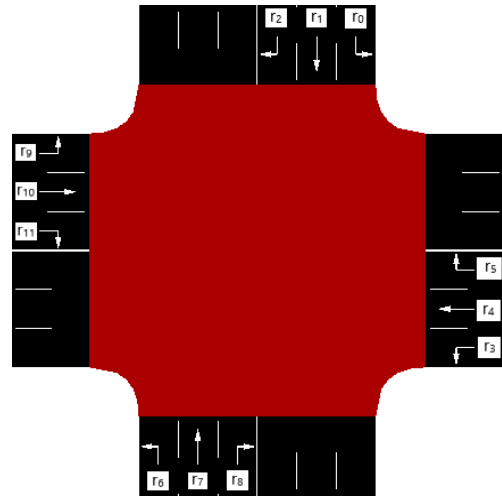


Fig. 1: 4X4 intersection featuring three dedicated incoming and outgoing lanes on each road.

B. Assumptions

The traffic situation has to be discussed before designing our AIM strategy. Thus we use the following assumptions for making necessary decisions that affect our definitions and algorithm:

- Once all vehicle overtaking and lane-switching maneuvers have been completed, vehicles driving in a lane must adhere to the predefined direction set by their respective routes.
- All vehicles in the system are fully autonomous, operating deterministically along specified lanes and speeds. This model excludes considerations for pedestrians, cyclists, or special privilege vehicles
- Vehicles communicate seamlessly upon entering each other's range, facilitated by the intersection's IM unit as a signal amplifier. This process minimizes delays and data loss, focusing solely on vehicle-to-everything (V2X) communication. We do not provide specific network modeling details.

- The adjusted speed and time estimations carried out by Connected Autonomous Vehicles (CAVs) are highly accurate. These estimations serve as the foundation for both local and global decisions made by Autonomous Vehicles (AVs) at the intersection.

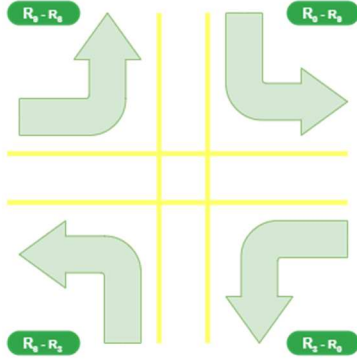


Fig. 2: The paths in Green have no restrictions. Yellow-coloured paths are algorithmically controlled.

C. Conflict matrix

We define conflicting routes as those that overlap within the traffic intersection area. The routes of vehicles can be determined based on the lanes they traverse. A conflict matrix, illustrated in Table I, represents these conflicts. Each row corresponds to available routes, and columns represent potential conflicts. Vehicles and the IM unit maintain this static conflict matrix for identifying potential conflicts as vehicles approach. If a vehicle approaches from a conflict-free lane, it can traverse safely at its chosen speed. Refer Fig. 2 Otherwise, it dynamically assesses real-time intersection occupancy using the Priority list.

TABLE I: Conflict Matrix

Route	Other routes that lead to collision
0	None
1	1, 4, 5, 8, 10
2	2, 5, 7, 10, 11
3	None
4	4, 1, 7, 8, 11
5	5, 1, 2, 8, 10
6	None
7	7, 2, 4, 10, 11
8	8, 1, 4, 5, 11
9	None
10	10, 1, 2, 5, 7
11	11, 2, 4, 7, 8
Total	77

D. Priority list

The design incorporates a priority list that consists of relevant vehicles and their associated information. The information for each vehicle includes:

- Vehicle unique ID: Identified by $i \in \{0, 1, 2, \dots\}$

- Vehicle route ID: Represented by route $i \in \{r_0, r_1, r_2, \dots, r_{11}\}$
- Estimated departure time of the vehicle from the intersection: Denoted as $t_i \in [0, 1)$

The IM efficiently manages system details, enhancing coordination and decision-making for autonomous vehicles within the intersection, ensuring smooth operation.

E. Traffic control strategy

The control strategy can be summarized as follows:

- Initial/Starting state: The priority list is empty, as there are no vehicles present in the system at the beginning.
- Once a vehicle enters the defined communication range of the intersection, it shares its basic information with all other vehicles. This information includes the vehicle's ID (denoted as i) and its route ID (denoted as $route_i$).
- To determine the vehicle's priority in the intersection, each vehicle first checks the conflict matrix. If there are no probable conflicts detected (typically seen with left-turning vehicles), the algorithm assigns a NULL or empty value to the vehicle's entry in the priority list. This signifies that the vehicle can safely pass through the intersection at its desired speed. The algorithm then proceeds to step (4).
- However, if there are potential conflicts for the vehicle's route (e.g., for $route_i = r_1$), the algorithm identifies the routes that could potentially conflict with $route_i$. Taking $route_i = r_1$ as an example, the potential conflict routes would be r_1, r_4, r_5, r_8 , and r_{10} .
- Each vehicle searches its priority list to identify registered vehicles for the current conflicting route. It then adjusts its speed and communicates t_{max} (maximum time to reach the intersection) to other vehicles. If no vehicles are registered on the current conflict route, t_{max} is set to NULL, signifying no conflicting vehicles.
- Vehicle i adjusts its speed based on feedback, aiming to arrive slightly after t_{max} if it is not NULL. CAVs have flexibility in speed adjustment methods, and all vehicles follow constant deceleration to reach newly calculated speeds. The goal is quick intersection traversal while respecting t_{max} and ensuring safe spacing. Vehicle i also shares its estimated arrival and departure time, t_i , for coordination with other vehicles.
- Vehicles update their priority lists with vehicle ID (i), route ID ($route_i$), and departure time (t_i). Upon completing the intersection passage, vehicles signal others for list updates. This ensures accurate priority information. Vehicles then compute local and global decisions, considering route priorities, departure times, conflicts, and intersection coordination.
- Repeat steps (2) to (5). The flowchart of the above steps is shown in Fig. 3, in which the steps inside the dashed box are local and global decisions. The pseudo-code of the algorithm performed by each vehicle is given in Algorithm 1.

Algorithm 1 Updating speed based on calculated intersection arrival time

Input: In the context of the conflict list/matrix (M), priority list (L), the approaching vehicle (vehicle i), its ID (i), and its route ID ($route_i$)

Output: New arrival time t_{max} on i 's current lanes

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1: if  $route_i$  has no conflicting lane in  $M$  then
2:    $t_{max} = \text{NULL}$ 
3: else
4:   for each conflict lane  $r_j$  in  $M$  do
5:     Get leaving time  $t_j$  on line  $r_j$  from  $L$ 
6:   end for
7:    $t_{max} = \max\{t_j\}$ 
8:   if  $steps = 100$  then
9:     for each vehicle  $j$  do
10:      Get leaving time  $t_j$ 
11:    end for
12:    Arrange  $L$  in ASC based on minimum chained
    vehicles
13:     $t_{max} = \max\{L\}$ 
14:  end if
15:  Go to step 4
16: end if

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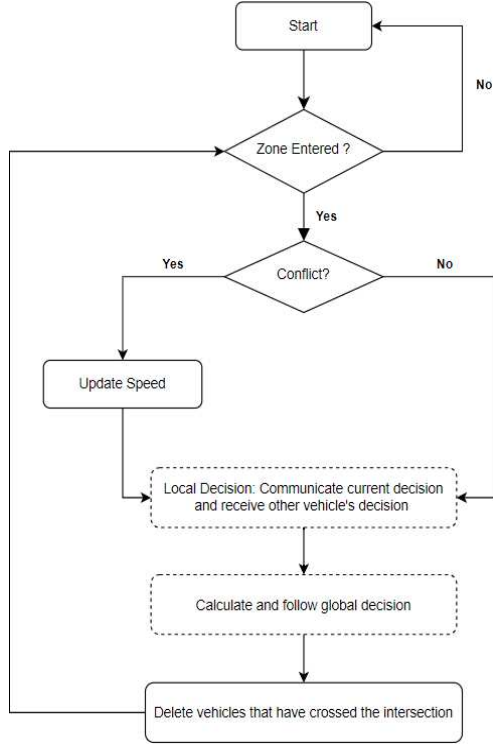


Fig. 3: Flowchart of the proposed IM strategy.

A. Simulations in SUMO

To validate our AIM (Autonomous Intersection Management) strategy, we employ the SUMO (Simulation of Urban MObility) software, an open-source traffic system simulation tool. Additionally, we use the Plexe API, an extension of SUMO for autonomous driving scenarios. Simulations consider the intersection environment in Fig. 1, where parameters and vehicle behaviors are configured to assess AIM strategy performance, including traffic flow, intersection optimization, and conflict reduction.

This simulation setup allows analysis of metrics like waiting times, throughput, resource efficiency, and overall system performance. Results provide quantitative data to validate our AIM strategy and compare it to existing approaches. Controlled experimentation under various traffic scenarios refines and enhances the AIM strategy for real-world autonomous driving environments. In the simulations, the following parameters are considered:

- Distance from starting points to stop line: It is set at 500 meters. This distance determines the initial positioning of the vehicles approaching the intersection.
- Distance from lane end to intersection center: Also called intersection radius, it is set at 15 meters. The overlaps are estimated to happen in this area.
- Location of the IM unit: The IM unit is placed at the center of the intersection, serving as the communication and amplification hub for the vehicles.
- IM unit Communication range: It is set to 400 meters. Vehicles can communicate with each other and the IM unit effectively within this range.
- Average/Default speed: It is set to 60 km/h, which is approximately 16.67 m/s. This speed determines the initial velocity of the vehicles approaching the intersection.
- Vehicle length: It is set to 4 meters. This parameter influences the spacing and positioning of vehicles within the intersection area.
- Step duration: It is set to 1 ms. This parameter dictates the frequency at which algorithm dictates environment parameters.
- Other parameters: The remaining parameters are set to their default values as provided by SUMO.

By configuring these parameters, simulations better replicate the intersection environment for evaluating the AIM strategy's real-world performance. In this AIM strategy, the speed adjustment method used is continuous and constant deceleration. The vehicle must arrive after $t_{max} + t_{\delta}$, where t_{max} is the maximum of the leaving times of the conflicting vehicles, and t_{δ} is a 100ms safety buffer.

For simulation, a straightforward speed adjustment method is employed. When the vehicle receives t_{max} , it checks if speed reduction is needed. If so, it adjusts speed based on the distance to the stop line (D_{int}), current time (t_c), and t_{δ} . i.e., $D_{int} / (t_{max} + t_{\delta})$ to be exact. This dynamic adjustment ensures a safe distance and arrival after $t_{max} + t_{\delta}$,

enhancing traffic efficiency. Note that this method is chosen for simulation and can be replaced with alternatives meeting the same arrival time requirement.

In our evaluation, we compare the AIM strategy with two other methods: traditional fixed-phase traffic lights and intelligent traffic lights. The traditional system uses a fixed three-phase signal timing scheme with a 120-second cycle, comprising of phases in the given order:

- Green light for going straight: This phase allows vehicles to proceed straight through the intersection. The duration is set to 30 seconds.
- Yellow light: The yellow light indicates the transition from the green phase to the next phase. It lasts for 5 seconds.
- Green light for right turn: This phase permits vehicles to make right turns at the intersection. The duration is set to 20 seconds.
- Yellow light: Similar to before, the yellow light provides a transition period between phases and lasts for 5 seconds.
- Red light: The red light signals a complete stop, staying active until the next green phase begins.

The fixed-phase timing scheme is a traditional traffic control approach widely used in intersections. Additionally, we evaluate intelligent traffic lights, which employ dynamic control methods based on real-time traffic conditions, and potentially advanced technologies that employ machine learning or sensors to optimize traffic flow.

By comparing AIM with these methods, we assess its effectiveness in terms of traffic efficiency, congestion reduction, and intersection management. In all the simulations, consistent traffic density is maintained across all lanes with vehicles spawned every 6 seconds at a 30% spawn probability for each lane. This ensures a balanced scenario for fair strategy evaluations. The results indicate AIM's superiority due to dynamic speed adjustments and real-time decision-making through vehicle communication. Compared to fixed traffic lights, AIM offers flexibility, allowing vehicles to pass efficiently and improve traffic flow.

In adaptive or intelligent lights systems, signal controller can extend or reduce phase duration based on traffic. AIM's superior performance suggests its adaptability surpasses the much advanced algorithms in intelligent systems. These results support AIM's effectiveness in enhancing traffic management and advancing autonomous driving.

B. A detailed discussion of results

All the durations are in seconds and the speed is in meters per second(m/s). First, we compare our method with the traditional intersection method, i.e., using traffic lights and the current trend of adaptive/intelligent signaling. The subfigures (a) of charts(Fig. 4, 5, and 6) compares vehicle acceleration patterns. The metric is acceleration VS time. All the metrics used here are available to plot in SUMO. The AIM strategy exhibits a narrower acceleration range compared to intelligent and traditional traffic lights. AIM's consistent acceleration sug-

gests smoother and more efficient vehicle movement, reducing sudden braking or deceleration.

The subfigures (b) of charts (Fig. 4, 5, and 6) examines speed deviations over time. The metric used here is speed VS time. AIM maintains positive and uniform speed deviations, ensuring stable vehicle speeds throughout the simulation. In contrast, traditional and intelligent traffic lights show non uniform deviations, sometimes reaching zero, indicating speed variations and potential stops. AIM's ability to maintain continuous speed contributes to improved traffic flow and reduced travel time compared to both traditional and intelligent traffic lights.

The subfigures (c) of charts (Fig. 4, 5, and 6) compares the metric: number of halting VS running vehicles over time. AIM has zero halting vehicles, while intelligent traffic lights have some, and traditional traffic lights have a higher count.

The subfigures (d) of charts (Fig. 4, 5, and 6) compares the duration distribution of vehicles. The metric used here is the average travel duration of vehicles from the start to the end of the simulation map (starting point of one lane and end point of any other lane) in Fig. 1. The duration is counted as the time a vehicle spends in the simulation i.e., from start to finish. In our strategy, durations of all vehicles are almost equal but with the traditional strategy and intelligent lights strategy it varies greatly. Some vehicles have less duration while others have more. Thus proving the efficiency of this approach.

Second, we compare our proposed system with three other methods as mentioned in [12], [13], and [14]. The paper [13] proposes a State Feedback Control (SFC) algorithm and the paper [14] proposes Multiagent System Intersection Management (MSIM). All these algorithms are simulated with the same settings as described in the System Proposal. The implementations [12], [13] that employ traffic lights, also are treated as systems with no limit to the number of inbound vehicles. A primitive comparison is shown in Table II.

TABLE II: Comparative Study

Strategy	t_{travel} (seconds)	V_{avg} (m/s)
SFC	35	22
MSIM	32	27
V2V-IM	29	28
PROPOSED METHOD	28	28

V. CONCLUSIONS AND FUTURE SCOPE

In conclusion, allowing multiple vehicles to share information and learn from each other in real-time, can improve the scalability, robustness, efficiency, and safety of the autonomous driving system. The proposed method's effectiveness over the traditional fixed and adaptive phase traffic light methods is backed by results. Various players within the autonomous driving sector may adopt distinct centralized approaches. At intersections, our decentralized algorithm can be adapted as an open standard, facilitating seamless communication among all vehicles in the environment. However, it is worth noting that the implementation of AIM strategies is

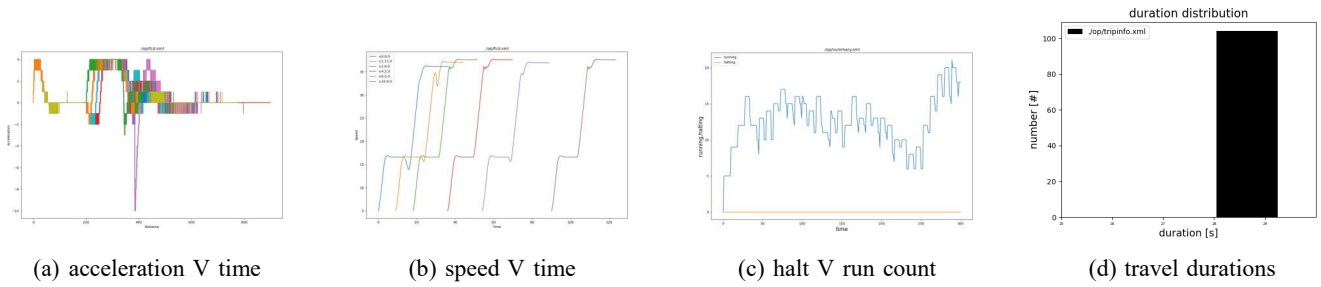


Fig. 4: Results for the proposed strategy

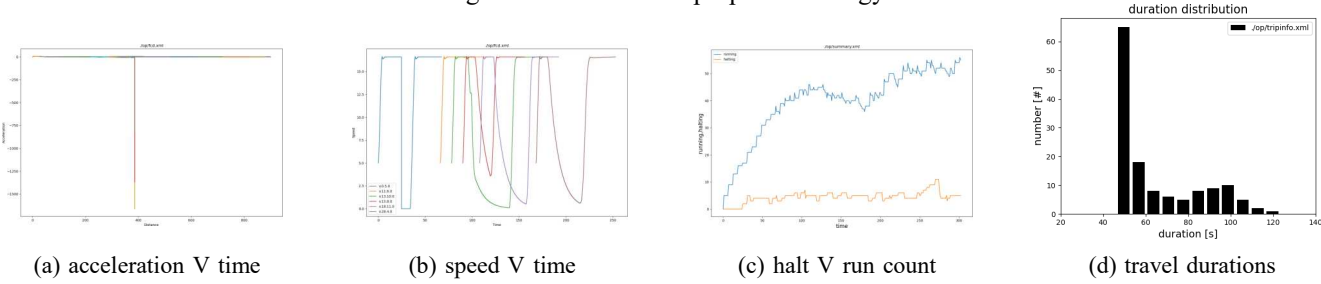


Fig. 5: Results for the intelligent intersection.

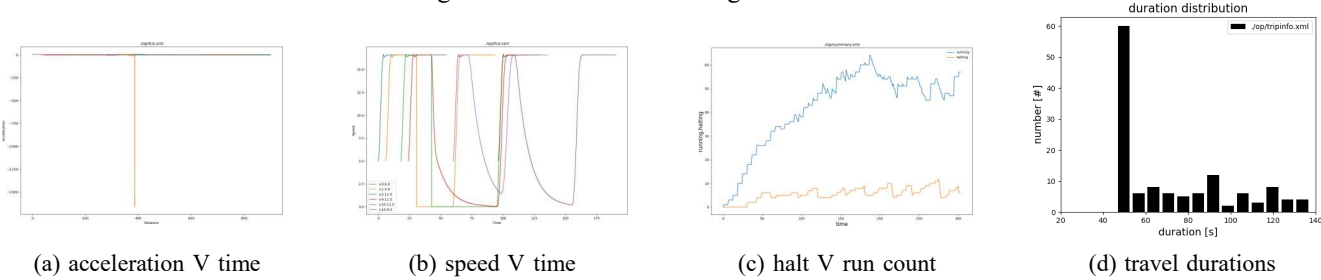


Fig. 6: Results for traditional traffic lights.

still an active area of research and some challenges need to be addressed such as communication reliability.

This approach is specifically designed for 4-way intersection. A generalized algorithm to cover different types of intersection can be developed. Other elements can be added to environment that affect decision making like emergency vehicles, pedestrians and blockades. The proposed system can be implemented on a Raspberry Pi or Arduino chip and simulated for training and evaluation purposes, and subsequently in real-time environment using actual vehicles in heavy-traffic intersections.

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